Ch. 7 Non-Textbook Questions

Principles of Microeconomics

July 15, 2026

1. Consider a market where supply is given by $Q_S = 10P - 40$ and demand is given by $Q_D = 200 - 10P$.

   (a) What is the market equilibrium price and quantity?

   (b) What is consumer surplus at the market equilibrium?

   (c) What is producer surplus at the market equilibrium?

   (d) If the government implements a price floor of \$15, what quantity of the good gets traded?

   (e) In the world of the price floor, what are consumer surplus and producer surplus?

   (f) In the world of the price floor, what is deadweight loss?

   (g) If the government instead implements a price ceiling of \$9, what quantity of the good gets traded?

   (h) In the world of the price ceiling, what are consumer surplus and producer surplus?

   (i) In the world of the price ceiling, what is deadweight loss?

   (j) If the government instead implements a tax of \$4 per good sold on suppliers, what price do consumers end up paying?

   (k) In the world of the tax, what price do suppliers end up receiving

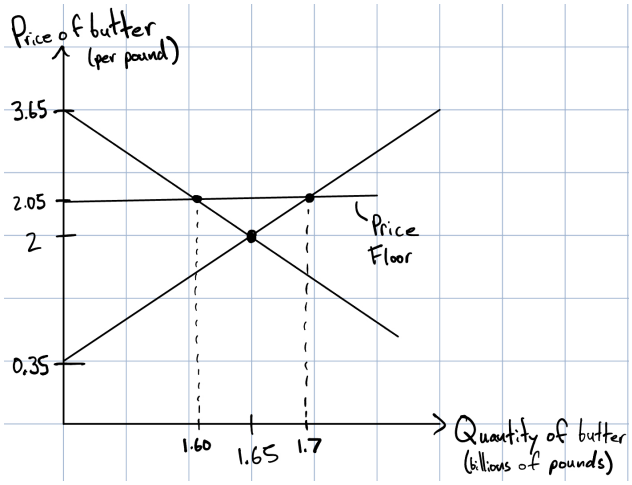
   (l) In the world of the tax, what quantity of the good gets traded?

   (m) In the world of the tax, what are consumer surplus, producer surplus, and tax revenue?

   (n) In the world of the tax, what is deadweight loss?

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2. The U.S. Department of Agriculture (USDA) administers a price floor for butter which the 2008 Farm Bill set at $2.05 per pound. At that price, the quantity of butter supplied in 2010 was 1.7 billion pounds. The quantity demanded was 1.6 billion pounds. To support dairy farmers, the USDA chose to buy up the 100 million pound surplus. The accompanying diagram illustrates the market for butter in 2010.



(a) In the absence of a price floor, how much consumer surplus would there have been? How much producer surplus? What would total surplus have been?

(b) What was consumer surplus with the price floor?

(c) What was producer surplus with the price floor? Recall that producers sold 1.7 billion pounds of butter – some to consumers and some to the USDA.

(d) How much money did the USDA spend buying up surplus butter?

(e) The USDA had to spend tax revenue to buy up the surplus butter. Therefore, to calculate total surplus for the market, we have to subtract the decrease in tax revenue. Using your answers from parts (b)-(d), what was total surplus in the market for butter? How does this compare to total surplus without a price floor?